CO3- ASSESSMENT TOOL 3

ERROR ANALYSIS TASK

DIGITAL SIGNAL PROCESSING | QUANTIZATION

Truncation vs. Rounding

Quantization Error in a 4-bit Quantizer

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Question & Definition

A 4-bit quantizer with input x = 0.73 V produces an output of 0.625 V using truncation, whereas the expected value is 0.75 V. Identify the implementation error and compute the correct quantization error.

Quantization

The process of converting a continuous input signal into the nearest discrete digital level.

Quantization Error

The difference between the input value and the quantized output.

$$\text{Quantization Error} = x - Q(x)$$

x = input value   Q(x) = quantized output

Step 1 & 2 — Given and the Error

What is given, and what went wrong

Step 1: Given

Input voltage, x

0.73 V

Output (truncation)

0.625 V

Expected output

0.75 V

Step 2: Identify the Error

- **The implementation uses truncation instead of rounding**
- Truncation always chooses the lower level
- Rounding chooses the nearest level
- So 0.625 V is incorrect and 0.75 V is the correct output

Step 3 — Correct Quantization Error

Substituting into $e = x - Q(x)$

- Error $= x - Q(x)$
- $= 0.73 - 0.75$
- **$= -0.02$ V**
- Magnitude, $|e| = 0.02$ V

$e = -0.02$ V

(magnitude 20 mV)

The negative sign shows the quantized level lies slightly above the input value.

Final Answer

Implementation Error	Truncation is used instead of rounding
Correct Quantized Output	$Q(x) = 0.75$ V
Correct Quantization Error	$e = -0.02$ V (magnitude = 0.02 V)

Conclusion: Rounding gives a more accurate output and reduces the quantization error.